

Decent Coaching Classes
VIII SB
Academic Conquest - VII

Subject: General Science (Ch. 1 to 6)

Answer Key | Marks: 30

Q.1 Choose the correct alternative. (1 mark each) [5]

1. (b) Bacteria
2. (c) Aedes aegypti
3. (b) Pascal (N/m²)
4. (c) 2 V
5. (c) Rutherford

Q.2 Fill in the blanks. (1 mark each) [4]

- (a) five
- (b) pancreas
- (c) non-contact
- (d) tin

Q.3 Do as directed. [6]

(A) True / False (1 mark each)

- (a) True.
- (b) False. Tuberculosis is an infectious disease (caused by Mycobacterium tuberculae).
- (c) False. A proton is a positively charged particle.

(B) Match the pairs (1 mark each)

1 → c (Plasmodium vivax) 2 → a (Vibrio cholerae) 3 → b (Entamoeba histolytica)

Q.4 Scientific reasons (any 2) (2 marks each) [4]

- (a) The bottom surface of a camel's feet is broad, so its weight is spread over a large area. Since pressure = force / area, the pressure on the sand becomes small. Hence the feet do not sink.
- (b) An atom has equal numbers of positively charged protons and negatively charged electrons. The total positive charge balances the total negative charge, so the net charge is zero.
- (c) Milk is a colloid. Its particles (about 10⁻⁵ m) are smaller than the pores of ordinary filter paper, so they pass through it along with the water and cannot be separated.

Q.5 Short answers (any 3) (2 marks each) [6]

- (a) Any two (1 mark each): Viruses are extremely minute (10–100 nm) and seen only with an electron microscope; a virus is a DNA/RNA molecule covered by a protein coat; they survive and multiply only inside living cells; they cause diseases in plants and animals; they are not considered fully living.
- (b) Any two points (1 mark each, with example):
Infectious: spread through contaminated air, water, food or vectors; caused by pathogens. E.g. dengue, cholera, TB.
Non-infectious: do not spread through infection or contact; arise in the body itself due to some reason. E.g. diabetes, heart disease, cancer.
- (c) Given: weight (force) = 150 N, area = 0.5 m². Pressure = Force / Area = 150 / 0.5 = 300 N/m² = 300 Pa. (Formula and substitution 1 mark; answer with unit 1 mark)
- (d) Ca O → valencies 2 2 → cross multiply: Ca₂O₂ → divide by 2 to get the smallest whole numbers: CaO. (Method 1 mark; formula 1 mark)

Q.6 Answer in detail (any 1) (5 marks) [5]

- (a) **Dry cell.** Labelled diagram (2 marks): zinc cover (negative terminal), carbon/graphite rod with cap (positive terminal), manganese dioxide paste, electrolyte (wet pulp of ZnCl₂ and NH₄Cl), divisor, protective cover.

Construction (1 mark): The outer container is a zinc layer (negative terminal). A graphite rod at the centre is the positive terminal. A paste of manganese dioxide is filled around the rod, and the space between is filled with the electrolyte, a wet pulp of zinc chloride and ammonium chloride.

Working and use (2 marks): Chemical reactions between these substances produce electric charge on the two terminals, creating a potential difference (about 1.5 V) so that current flows through the circuit. The reaction is slow, so large current cannot be obtained, but the shelf life is long. Used in radios, wall clocks and torches.

(b) Rutherford's experiment. Experiment (1 mark): A very thin gold foil (about 10^{-4} mm) was bombarded with positively charged α -particles from a radioactive source; their paths were observed on a fluorescent screen around the foil.

Observations (2 marks): Most α -particles passed straight through the foil; a small number were deflected through small angles; a still smaller number were deflected through large angles; about 1 in 20000 bounced back.

Conclusions (2 marks): Most of the atom is empty space. A positively charged, heavy nucleus is present at the centre and contains almost all the mass; electrons revolve around it.